

Smart Queue Management System For Government Offices In Nepal

BY:

Siddhartha Shakya

T.U REGISTRATION NO: 7-2-551-88-2023

ROLL NO: 16055/23

NATIONAL COLLEGE OF COMPUTER STUDIES

A Project Proposal Submitted To

Faculty of Management, Tribhuvan University

in partial fulfillment of the requirements for the degree of

Bachelor of Information Management (BIM)

Paknajol, Kathmandu

August, 2026

1. Introduction

1.1 Introduction

Government offices in Nepal provide a wide range of essential public services, including citizenship registration, passport-related services, driving license processing, recommendation letters, and other administrative services. As the demand for these services continues to increase, many government offices face challenges in managing large numbers of service seekers efficiently. Long waiting times, overcrowded waiting areas, manual token distribution, and limited visibility of queue progress often result in inconvenience for citizens and increased workload for government staff.

Recent advancements in digital technologies have created opportunities to modernize public service delivery through smart queue management solutions. A digital queue management system enables citizens to obtain queue tokens electronically, monitor their queue status in real time, and receive timely updates without having to remain in a physical queue throughout the waiting period. At the same time, it provides government staff with efficient tools to manage queues, monitor service counters, and maintain organized service delivery.

This project proposes the development of a **Smart Queue Management System for Government Offices in Nepal**, implemented as a Progressive Web Application (PWA). The system is designed to support multiple government offices, multiple services, and multiple service counters within a centralized platform. Citizens will be able to generate same-day digital queue tokens either as guest users or registered users, complete their arrival through QR-based or manual check-in, and track their queue position and estimated waiting time in real time. The system also includes features such as priority queue management for exceptional cases, dynamic waiting time estimation, queue audit logging, and a staff dashboard for efficient queue administration.

The proposed system aims to improve the efficiency, transparency, and fairness of queue management in government offices while reducing unnecessary waiting and overcrowding. By replacing conventional manual queue management with a digital solution, the project seeks to enhance the overall experience of citizens and support government staff in delivering public services more effectively.

1.2 Problem Statement

Many government offices in Nepal continue to rely on conventional methods of queue management, where citizens are required to wait in long physical queues to access public services. In many cases, queue handling is performed manually or through basic token systems that provide limited information about service progress. As a result, citizens often experience uncertainty regarding their waiting time and are required to remain at the service location until their turn arrives. This not only leads to overcrowded waiting areas but also creates an inconvenient and time-consuming experience for service seekers.

The absence of a transparent and efficient queue management mechanism also makes it difficult for government staff to monitor queue operations effectively, particularly when multiple services and service counters are involved. Citizens have limited visibility in the current queue status, while staff must manage queues manually under varying service conditions, increasing the possibility of operational inefficiencies and inconsistent queue handling.

Therefore, there is a need for a digital queue management system that provides transparent queue tracking, efficient queue administration, and real-time information for both citizens and government staff. Such a system can improve the overall efficiency of service delivery while enhancing the experience of citizens seeking government services in Nepal.

1.3 Objectives

- To design and develop a Smart Queue Management System for government offices in Nepal that improves the efficiency and transparency of queue management through digital queue administration and real-time queue tracking.

1.4 Development Methodology

The development of the **Smart Queue Management System for Government Offices in Nepal** will follow the **Incremental Waterfall Model**. This methodology was selected because it provides a structured and systematic approach to software development while allowing the system to be developed and tested in manageable increments. As the project is being developed by a single developer within a limited academic timeframe, the Incremental Waterfall Model offers a practical balance between disciplined planning and gradual implementation.

In this approach, each development phase is completed before proceeding to the next phase within an increment. The project begins with requirement analysis and system design, followed by implementation, testing, and documentation. After completing one increment, the next set of features is developed using the same sequence until the entire system is completed. This approach enables continuous evaluation of completed modules while reducing the complexity of implementing the entire system at once.

The development process for this project is planned in the following increments:

- **Increment 1:** Requirement identification, feasibility study, system analysis, database design, and user interface design.
- **Increment 2:** Development of user authentication, government office management, service management, and service counter management modules.
- **Increment 3:** Implementation of digital token generation, QR-based and manual check-in, queue management, and staff dashboard functionalities.
- **Increment 4:** Development of real-time queue tracking, waiting time estimation, priority queue handling, queue audit logging, and basic reporting features.
- **Increment 5:** System integration, testing, debugging, deployment, and project documentation.

By adopting the Incremental Waterfall Model, the project will be developed in a controlled and organized manner, ensuring that each functional module is completed, tested, and integrated before progressing to subsequent increments. This approach supports the delivery of a reliable and maintainable queue management system within the project's scope and timeline.

2. Requirement Identification and Feasibility Study

2.1 Requirement Identification

The requirements of the proposed **Smart Queue Management System for Government Offices in Nepal** were identified by analyzing existing queue management challenges and considering the needs of citizens and government staff. The requirements are categorized into functional and non-functional requirements.

2.1.1 Functional Requirements

- **User Management**

The system shall support registered citizens, government staff, and administrators with appropriate access permissions. Registered users can access additional features such as queue history.

- **Government Office and Service Management**

The system shall support multiple government offices, services, and service counters through a centralized platform while maintaining independent queue operations for each office.

- **Digital Token Generation**

The system shall allow citizens to select a government office and service to generate a same-day digital queue token with a unique token number and QR code.

- **Queue Check-in Management**

The system shall provide QR-based and manual check-in options to confirm citizen arrival before joining the active queue.

- **Queue Management and Counter Handling**

The system shall allow staff to manage queues by calling tokens, assigning counters, skipping unavailable citizens, recalling tokens, and completing services.

- **Real-Time Queue Tracking and Waiting Time Estimation**

The system shall provide citizens with real-time queue status, position updates, and estimated waiting time based on current queue conditions.

- **Priority Queue Management**

The system shall support verified priority handling for cases such as senior citizens, persons with disabilities, pregnant women, and emergencies while maintaining transparency through recorded approvals.

- **Queue Audit Logging**

The system shall record important queue activities, including token generation, check-in, priority approval, queue changes, and service completion for accountability.

- **Reporting and Monitoring**

The system shall generate basic reports related to service activities, token usage, and queue performance.

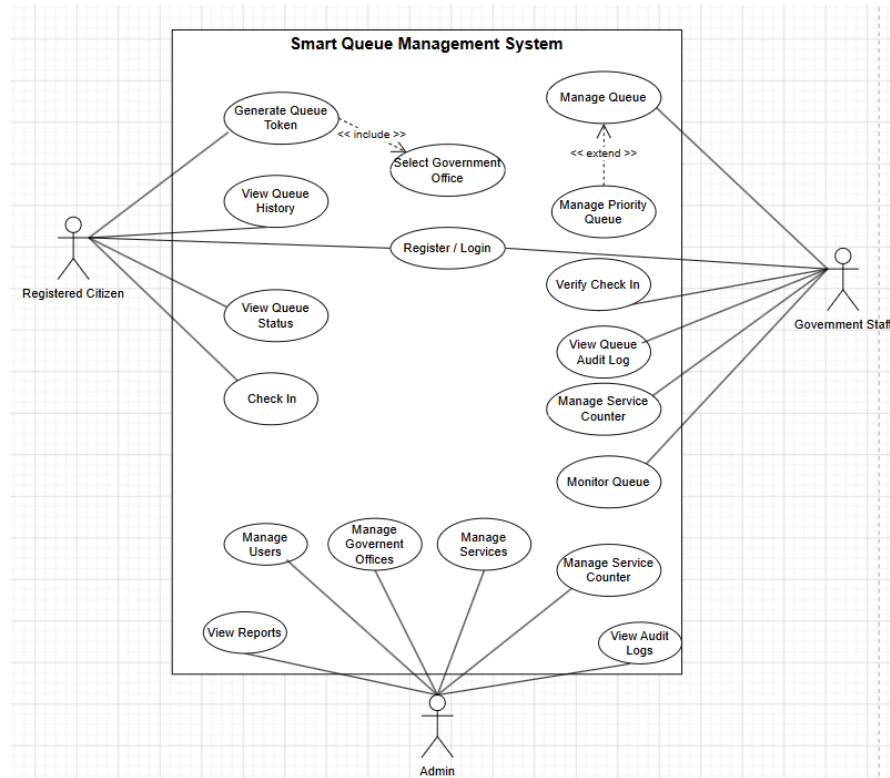


Figure 2.1: Use Case Diagram

2.1.2 Non-Functional Requirements

- **Usability**

The system should provide a simple and user-friendly interface for citizens and staff.

- **Performance**

The system should provide fast responses and real-time queue updates.

- **Security**

The system should ensure secure authentication, authorization, and protection of user data.

- **Scalability**

The system should support additional government offices, services, and counters in the future.

- **Reliability**

The system should maintain accurate queue information and operate consistently during service hours.

- **Accessibility**

The system should be accessible through mobile and desktop devices using a Progressive Web Application (PWA).

2.1.3 Feasibility Study

a) Technical Feasibility

The system is technically feasible as it will use modern web technologies such as React.js, Node.js, Express.js, PostgreSQL, and Socket.IO. These technologies provide the required capabilities for real-time queue management, authentication, and QR-based check-in.

b) Operational Feasibility

The system is operationally feasible as it improves existing queue management processes without requiring major changes to government office workflows. Both QR-based and manual check-in methods ensure accessibility for citizens.

c) Economic Feasibility

The system is economically feasible as it will be developed using open-source technologies, requiring minimal financial investment for development and testing.

2.2 Related Work/Literature Review

Queue management has been widely studied in service operations because waiting time, queue length, service capacity, and customer expectations directly influence service efficiency and user experience. Traditional physical queues require customers to remain at the service location until their turn, whereas virtual queue systems allow customers to join queues remotely and monitor their progress digitally. Hu et al. (2021) examined virtual queues and found that providing customers with information about their queue position and remaining waiting time can influence their perception of the waiting experience.

Waiting-time information is particularly important in a digital queue management system. Antonides et al. (2002) found that providing information about expected waiting times can reduce customers' overestimation of how long they have been waiting. Similarly, Yu et al. (2022) examined wait-time information in a large-scale virtual queue and found that both the initial waiting-time estimate, and subsequent updates influenced customer abandonment behavior. These findings support the inclusion of real-time queue status and dynamic waiting-time estimation in the proposed system.

Queue fairness and priority handling are also important considerations in service systems. Yu et al. (2018) studied service systems in which customers have different priorities and examined how delay information can be used while managing different customer types. Their findings demonstrate the importance of balancing service priorities with customer expectations. Batt and Terwiesch (2015) further found that customers are sensitive to changes in observable queue conditions, including situations where other customers appear to move ahead in the queue. These findings are relevant to the proposed system's priority management and queue transparency features.

Recent developments in web-based queue management have also demonstrated the practical use of digital tokens, online queue registration, real-time status tracking, and QR-based interactions. Fauzi and Suhaimi (2025), through the development of QKlinik, demonstrated a web-based queue management system incorporating online queue registration, real-time status tracking, and QR-code functionality. Similarly, Kousheel et al. (2026) proposed a web-based smart queue and token management system for service organizations, demonstrating the use of digital tokens and administrative queue control.

The reviewed literature indicates that digital queue management can improve the organization of service queues by providing users with queue information, reducing uncertainty during waiting, and supporting staff in managing service operations. However, much of the existing research focuses on commercial services, healthcare environments, or general-purpose queue systems. The proposed project focuses specifically on **government offices in Nepal** and combines same-day digital token generation, QR and manual check-in, multiple service counters, real-time queue monitoring, waiting-time estimation, controlled priority handling, and queue audit logging within a single platform. This provides a context-specific approach to applying digital queue management to government service delivery in Nepal.

2.3 Analysis and Design Tools

The **Smart Queue Management System for Government Offices in Nepal** will be analyzed and designed using a structured approach. The system will be divided into different processes and functions to clearly understand how queue-related data is collected, processed, stored, and transferred between citizens, government staff, administrators, and system components. This approach will help define the system requirements and establish an organized system workflow.

i. Analysis Tools

The following tools and techniques will be used during the analysis phase:

- **Use Case Diagram:** This will show how citizens, government staff, and administrators interact with the system and identify the major functions available to each user.
- **Requirement Specification:** Functional and non-functional requirements will be documented to define the services and quality standards expected from the system.
- **Feasibility Analysis:** The proposed system will be evaluated based on technical, operational, economic, and schedule feasibility to determine its practicality.
- **Process Analysis:** Major processes such as token generation, check-in, queue management, priority handling, counter assignment, and service completion will be analyzed to understand their inputs, processing steps, and outputs.

ii. Design Tools

The following tools and techniques will be used during the system design phase:

- **Draw.io (diagrams.net):** It will be used to create:
 - **Data Flow Diagrams (DFD):** To represent the movement of data between users, system processes, database, and external components.
 - **System Architecture Diagrams:** To represent the overall structure and interaction of the PWA, backend, database, and real-time communication components.
 - **Use Case Diagrams:** To illustrate interactions between system users and the major system functions.
 - **Activity and Sequence Diagrams:** To represent important system workflows and interactions between system components.
- **Figma:** It will be used to design user interfaces and create prototypes for citizens, government staff, and administrators before implementation.
- **Entity-Relationship Diagram (ERD):** The ERD will be used to design the PostgreSQL database by identifying entities such as users, government offices, services, service counters, tokens, queue records, priority requests, and audit logs, along with their relationships.

2.4 Implementation Tools

The Smart Queue Management System will be implemented as a **Progressive Web Application (PWA)** using modern web technologies. The selected tools and technologies will support the development of the user interface, server-side functionality, database management, real-time queue updates, and QR-based check-in.

i. Front-End Tools

The following technologies will be used for the front-end development:

- **React.js:** It will be used to develop the interactive user interfaces for citizens, government staff, and administrators.
- **Tailwind CSS:** It will be used to create responsive and consistent user interfaces across different screen sizes.
- **Vite:** It will be used as the development and build tool for the React application.
- **Progressive Web App (PWA) Technologies:** Service workers, web app manifests, and related PWA technologies will be used to enable installation and convenient access from mobile and desktop devices.

ii. Back-End Tools

The following technologies will be used for back-end development:

- **Node.js:** It will provide the server-side runtime environment for the application.
- **Express.js:** It will be used to develop RESTful APIs and handle communication between the front end, database, and other system components.
- **Socket.IO:** It will be used to provide real-time communication for queue status, token updates, service counter changes, and estimated waiting time updates.

iii. Database Tools

- **PostgreSQL:** It will be used as the primary relational database for storing user information, government offices, services, service counters, tokens, queue records, priority requests, and audit logs.
- **Prisma ORM:** It will be used to interact with the PostgreSQL database and manage database schemas and queries efficiently.

iv. Additional Tools

- **QR Code Library:** A suitable QR code library will be used to generate and validate QR codes for the citizen check-in process.
- **Git and GitHub:** They will be used for source code management and maintaining the project repository.
- **Visual Studio Code:** It will be used as the primary development environment for writing and managing the application code.

3. Expected Outcome

1. Generate Queue Token

2. View Queue Status

3. Staff – Manage Queue (Call Next Token)

Position	Token No.	Service	Status	Checked-in
1	TRF-017	License Renewal	Waiting	Yes
2	TRF-018	License Renewal	Waiting	Yes
3	TRF-019	License Renewal	Waiting	No
4	TRF-020	License Renewal	Waiting	Yes

Figure 3.1: Some Core Features

The proposed Smart Queue Management System is expected to provide a digital and organized approach to managing queues in government offices in Nepal. The system is expected to achieve the following outcomes:

- **Reduced Physical Waiting:** Citizens will be able to obtain same-day digital queue tokens and monitor their queue progress without continuously waiting in a physical queue.
- **Improved Queue Transparency:** Citizens will be able to view their queue position, current service status, and estimated waiting time, providing greater visibility in the service process.
- **Efficient Queue Administration:** Government staff will be able to manage multiple queues and service counters through a centralized dashboard, improving the organization of daily service operations and reducing the crowd in waiting areas.
- **Improved Accountability:** Queue audit logs will provide a record of important queue activities, support transparency and allow authorized staff to review queue operations.

Overall, the project is expected to demonstrate how a digital queue management system can improve the **efficiency and transparency of government service queue management** while providing a more convenient experience for citizens.

4. Project Schedule

The **Smart Queue Management System for Government Offices in Nepal** is planned to be completed within a period of approximately **16 weeks (3–4 months)** following the Incremental Waterfall Model. The project schedule is divided into five development increments, with each increment focusing on a specific set of system functionalities. The activities are arranged sequentially so that the requirements and design of each increment are completed before its implementation, followed by testing and integration. This approach allows the system to be developed progressively while ensuring that each completed module is evaluated before moving to the next stage.

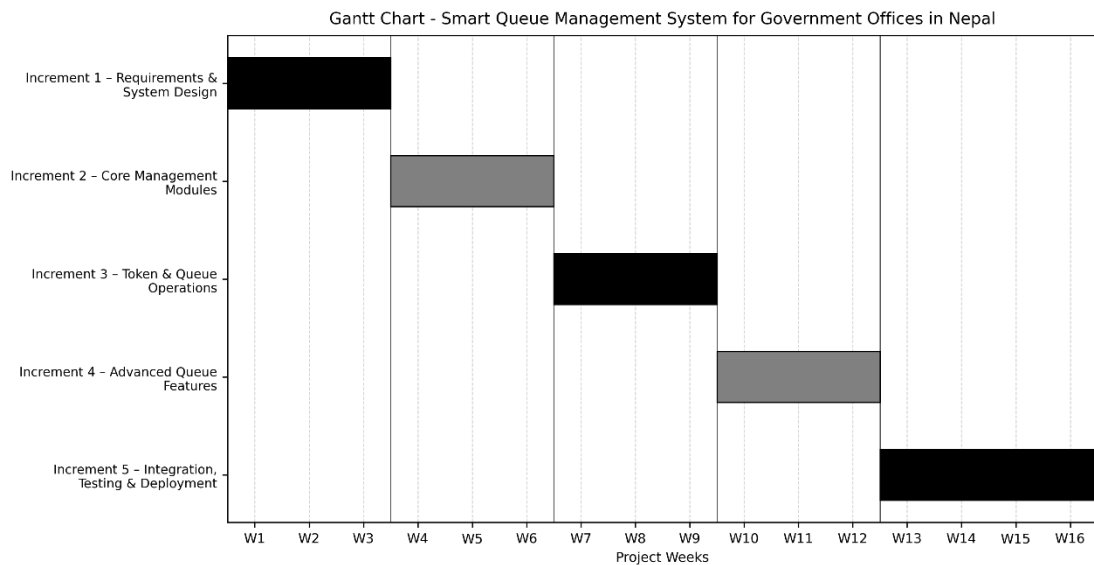


Figure 4.1: Gantt Chart

5. References

- Antonides, G., Verhoef, P. C., & van Aalst, M. (2002). Consumer perception and evaluation of waiting time: A field experiment. *Journal of Consumer Psychology*, 12(3), 193–202. https://doi.org/10.1207/S15327663JCP1203_02
- Batt, R. J., & Terwiesch, C. (2015). Waiting patiently: An empirical study of queue abandonment in an emergency department. *Management Science*, 61(1), 39–59. <https://doi.org/10.1287/mnsc.2014.2058>
- Fauzi, N. A. b. M., & Suhaimi, N. H. S. (2025). Design and evaluation of QKlinik: A web-based queue management system for outpatient clinics. *International Journal of Software Engineering and Computer Systems*, 11(2). <https://doi.org/10.15282/ijsecs.11.2.2025.12.0144>
- Hu, K., Xu, X., & Qu, A. (2021). *The psychology of virtual queue: When waiting feels less like waiting* [Research paper]. SSRN. <https://doi.org/10.2139/ssrn.3933040>
- Kousheel, M., Reddy, S. M., Deepak, K., Chary, M. S., & Rani, P. S. (2026). Smart queue and token management system. *International Journal of Innovative Research in Technology*, 12(10), 8041–8047.
- Yu, Q., Allon, G., Bassamboo, A., & Iravani, S. M. R. (2018). Managing customer expectations and priorities in service systems. *Management Science*, 64(8), 3942–3970. <https://doi.org/10.1287/mnsc.2017.2785>
- Yu, Q., Zhang, Y., & Zhou, Y.-P. (2022). Delay information in virtual queues: A large-scale field experiment on a major ride-sharing platform. *Management Science*, 68(8), 5745–5757. <https://doi.org/10.1287/mnsc.2022.4448>